Hannah T. Nguyen

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EDUCATION

The University of Tulsa — CyberCorps
B.S. Electrical and Computer Engineering — May 2027 (expected), GPA: 4.00
Minors in Mathematics and Computer Science

WORK EXPERIENCE

University School — **Tulsa, OK**
Lab Assistant — August 2025 - Present

- Provided classroom support and guidance for middle school Computer Science/Engineering courses by engaging students in hands-on projects.
- Managed system administration and configuration for a multi-node Raspberry Pi server rack.

U.S. Government — **Washington D.C.**
Software Developer Intern — Summer 2025

- Developed and designed an application to assist with data triage of a collection with the utilization of AI.
- Specialized in engineering, development, integration, and product evaluation services for the workgroup platform.
- Provided critical support for the infrastructure to allow for communications and collaboration.

USSS National Computer Forensics Laboratory — **Tulsa, OK**
Undergraduate Researcher — November 2023 - May 2025

- Conducted research on alternative reality software and tools, evaluating 5+ platforms to rapidly and cost-effectively recreate digital crime scenes.
- Developed low-cost methods for reconstructing large areas digitally to improve analysis and planning of protection zones during major events.

TURC Program — **Tulsa, OK**
Volunteer Java Programming Instructor — Summer 2024

- Co-taught two Java programming classes to 37 incoming freshmen.
- Gave the students a head start in their computer science education, enabling them to bypass the first two major computer science classes and take advanced courses during their first semester in college.

SKILLS & PROFICIENCY

- **Professional Skills:** Leadership & team building, time management, project management, technical writing & review, product testing, independent research, Vietnamese
- **Technical:** Hardware reverse engineering, Raspberry Pi, Arduino, STM32, electrical wiring, circuit analysis, mathematics, physics, soldering, multimeter
- **Software:** Arduino IDE, MathWorks (MATLAB, Simulink, Simscape), SOLIDWORKS, Fusion 360, Autodesk Inventor, Python, Java, C, C++, Lua, Git Bash, HTML, CSS, JavaScipt, Ubuntu, Linux, Shell Scripting, Windows, iOS, Visual Studio Code, LaTeX, Microsoft Office 365, Google Professional Suite, Altium Designer, KiCAD, VHDL, MIPS, Xilinx, Vue3, Vuetify3.

PROJECTS AND ACTIVITIES

Certifications

SANS GIAC Incident Handler Certification (GCIH) — January 2025

SANS GIAC Global Industrial Cyber Security Professional Certification (GCISP) — November 2025

TS/SCI Clearance

Projects

Personal Portfolio Website

- Created, designed, and deployed a personal portfolio website using 4+ front end languages and frameworks.
- Implemented responsive layouts, interactive features, and GitHub version control to showcase projects and skills.

Reverse Engineer PCB into FGPA — Spring 2025

- Reverse engineered a custom PCB and implemented the functionality on an FGPA board using VHDL.

University RFID Analysis — Fall 2023

- Conducted a capstone research project on reverse engineering TU ID card systems, identifying 3+ software and hardware vulnerabilities.
- Proposed and documented security solutions including password protection with lock bits, reinforced hardware design, and card number obfuscation.

Single-Port Read/Write Memory Design — Spring 2025

- Lab focused on designing and constructing a 16×4-bit memory device using a 64-bit shift register, address counter, and supporting logic.
- Implemented address comparison with XNOR/AND gates, clock division using T flip-flops, and read/write functionality through multiplexers and D flip-flops.
- Verified functionality with LEDs and 7-segment displays, and debugged timing and hardware issues to ensure reliable operation.

Projectile Trajectory Simulation — Spring 2025

- Project involved programming in MARS MIPS assembly and C to calculate projectile time-of-flight, maximum height, and trajectory angle based on user input.
- Applied physics equations of motion to model artillery firing scenarios, validated results across both implementations, and documented methodology, findings, and error analysis in a professional report.

IPS Display Module PCB — Fall 2025

- Designed an IPS Display module using an STM32G0B1CEU6 MCU and ER-TFT0.99-2 in Altium Designer.

Activities

Institute of Electrical and Electronics Engineers (IEEE) Chapter President and Honor Society Member — May 2024 - Present

- Led professional development workshops on practical skills like resume writing and soldering.
- Secured participation from a wide range of industry professionals for networking events.
- Increased the ECE students industry internship opportunities and participation by 300%.

Awards

Alan Paller Cyber Scholarship — January 2024 - Present

- Prestigious scholarship for excellence in cyber studies and research, and a commitment to serve with the federal government or OPM-approved organizations.
- The scholarship provides a stipend and the opportunity to take SANS courses and certification exams at no cost.

University of Tulsa Presential Scholarship — August 2023 - Present

- Full tuition scholarship awarded to high-achieving students who have demonstrated academic excellence in high school.

Tulsa Engineering Scholarship — May 2023 - Present

- Scholarship awarded to high-achieving students who major in an engineering field.

Oklahoma Academic Scholar — May 2023

- Honor accorded to students who accumulate a minimum grade point average of 3.7 or were in the top 10% of their graduating class and obtained a minimum of 27 ACT composite score or a 1220 SAT score.

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FTC State Championship Control Award Sponsored by ARM, Inc. — February 2023

- Awarded to high school teams that use sensors and software to increase robot functionality.
- Teams must demonstrate innovative thinking to solve challenges such as autonomous operation and improving mechanical systems using sensors and intelligent control schemes.